Residual Spectral–Spatial Attention Network for Hyperspectral Image Classification

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***Abstract*— In the last five years, deep learning has been introduced to tackle the hyperspectral image (HSI) classification and demonstrated good performance. In particular, the convolutional neural network (CNN)-based methods for HSI classification have made great progress. However, due to the high dimensionality of HSI and equal treatment of all bands, the performance of these methods is hampered by learning features from useless bands for classification. Moreover, for patchwise-based CNN models, equal treatment of spatial information from the pixel-centered neighborhood also hinders the performance of these methods. In this article, we propose an end-to-end residual spectral–spatial attention network (RSSAN) for HSI classification. The RSSAN takes raw 3-D cubes as input data without additional feature engineering. First, a spectral attention module is designed for spectral band selection from raw input data by emphasizing useful bands for classification and suppressing useless bands. Then, a spatial attention module is designed for the adaptive selection of spatial information by emphasizing pixels from the same class as the center pixel or those are useful for classification in the pixel-centered neighborhood and suppressing those from a different class or useless. Second, two attention modules are also used in the following CNN for adaptive feature refinement in spectral–spatial feature learning. Third, a sequential spectral–spatial attention module is embedded into a residual block to avoid overfitting and accelerate the training of the proposed model. Experimental studies demonstrate that the RSSAN achieved superior classification accuracy compared with the state of the art on three HSI data sets: Indian Pines (IN), University of Pavia (UP), and Kennedy Space Center (KSC).**

***Index Terms*— Attention network, convolutional neural networks (CNNs), hyperspectral image (HSI) classification, residual spectral–spatial attention network (RSSAN), spatial attention, spectral attention.**

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I. Introduction

HYPERSPECTRAL image (HSI) is full of redundantly continuous spectral and spatial information with high spatial resolution and correlation [1], [2]. Due to its rich information, HSI has been applied to many applications, such as urban development [3], surveillance [4], and environment management [5]. Hyperspectral image classification (HSIC) is one of the tasks to analyze HSI, which aims to classify each pixel in a HSI to one of a given set of land-cover classes, such as land and water [6].

Many methods have been proposed for HSIC. According to the feature learning manner, they can be roughly categorized to supervised feature learning [7], [8], unsupervised feature learning [9]–[12], and semi-supervised learning [13]–[15]. Supervised methods learn features only from a small amount of labeled data. Unsupervised methods aim to make full use of amounts of unlabeled data. Supervised methods are easy to meet overfitting if a complex model with a lot of trainable parameters is used. While unsupervised methods lack discriminative information. Semi-supervised methods are proposed to deal with the abovementioned problem by combining supervised feature learning of labeled data and unsupervised feature learning of unlabeled data. In this article, we only consider the case of supervised learning.

HSI often has a high spectral resolution. There is a lot of redundant information and a strong correlation between different spectral bands. Moreover, the high spectral resolution also causes the curse of dimensionality and requires more heavy computational effort. To avoid the curse of dimensionality, two methods including feature selection and feature extraction are used to reduce the dimension of spectral bands. Feature selection is to select some of the spectral bands that are more useful for the following task, such as band selection. Some band selection methods [16]–[20] are proposed to reduce the redundancy between different spectral bands to solve the abovementioned two problems. The abovementioned methods use the same subset of spectral bands for different

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classes. However, each spectral band should have different contributions to classify the pixels from different classes. We argue that each class should have its owner optimal subset of spectral bands, and these subsets for different classes may be different. For example, bands 1, 5, 10, … are useful to classify class 1, while bands 3, 7, 9, … are useful to classify class 2. Feature extraction is to combine these spectral bands together by some linear or nonlinear transformation. Traditional dimension reduction methods include principal component analysis (PCA) [21], independent component analysis (ICA) [22], and Fisher's linear discriminant analysis (LDA) [23]. PCA is applied to the spectral vectors and extracts the first k principal components as spectral features [21]. However, it needs manually setting the parameter k with expert experience because proper k will keep all useful information in spectral bands; otherwise, it may lose some useful information. As an unsupervised method, the first k principal components lack the discriminative information.

Spatial information is useful for HSIC [24]–[26]. To use the spatial information, many methods [27]–[29] use a pixel centric 3-D patch as the input to represent a pixel. It assumes that all the pixels in the neighborhood have the same contribution to classify the center pixel, also named spatial homogeneity. This assumption is true for most pixels in the homogeneous regions, while it may be not true for pixels that locate on the boundaries between two different classes. However, these methods do not distinguish contributions from different pixels in the neighborhood on classification. Meanwhile, not all the pixels in the neighborhood are useful to classify the center pixel; some of them may be harmful [30]. To alleviate the above problem, superpixel-based methods [31]–[35] propose to adopt the superpixel as the input. These methods first use a superpixel segmentation algorithm to segment the whole image to a lot of superpixels where pixels in a superpixel have the same class label if the proper number of superpixels is used. Then, some features are extracted from each superpixel as the input of the following model. An adaptive nonlocal spatial–spectral kernel is proposed to adaptively measure the similarities of different homogeneous patches in kernel feature space [36]. However, these methods need to select proper superpixel segmentation algorithm and hyperparameters, and superpixel feature extraction method. Moreover, patch size or the granularity of each superpixel has a great influence on the classification performance. Each pixel in the patch except the center pixel (named surrounding pixel) should have different contributions on the center pixel.

The disadvantages of traditional feature extraction methods are the requirement of handcrafted feature and their dependence on the prior information of the designers or domain experts, which results in a poor adaptation and generalization to different or unseen data sets. Besides, these handcrafted features are usually too low level and shallow to handle complex and new situations. As deep learning can extract high-level and more abstract features [37], they have been explored to deal with HSIC [38]–[40]. These methods include stacked autoencoders (SAEs) [9], [41]–[44], deep belief networks (DBNs) [10], [45], convolutional neural networks (CNNs) [7], [8], [46], recurrent neural networks (RNNs) [30], [47], and generative adversarial networks (GANs) [48]–[51]. First, unsupervised models, such as SAEs and DBNs, are introduced into HSIC. Chen *et al.* [9] used a multilayer SAE to extract deep features for HSIC for the first time. After the pretraining stage, a logistic regression classifier is added to the last layer of deep networks to fine-tune the whole network with labeled samples for HSIC. Similarly, Li *et al.* [10] used a single restricted Boltzmann machine (RBM) and a multilayer DBN to extract spectral–spatial features for HSIC. The main problem of SAE- and DBN-based methods is that the spatial information is not considered in the spatial feature extraction stage, as image patches are flattened in vectors. In order to alleviate the above-mentioned problem, the CNN-based methods are introduced into for HSIC [7]. As we know, CNN can learn features that keep the spatial relationship of the input data. However, these CNN models use the three principal components of raw HSI patches, which results to lose some spectral–spatial information. Moreover, Zhang *et al.* [8] proposed a spectral–spatial residual network (SSRN) that uses the spectral and spatial residual blocks consecutively to learn deep discriminative features from abundant spectral features and spatial contexts of HSI. However, in terms of spatial features, these deep learning-based methods use raw 3-D patch or spatial features after reducing the spectral feature dimensionality, ignoring the relationship between the neighboring pixels.

Although a lot of CNN-based models have been proposed for HSIC, there are still some problems that have not been well solved until now.

1) Do all bands provide the same contribution on a CNN for HSIC? If no, how to data-driven adaptively find informative bands that are useful for HSIC and implement it in an end-to-end network?
2) Do spatial information from all pixels in the neighborhood contribute equally to a CNN for HSIC tasks? If no, how to data-driven adaptively find informative pixels that are useful for HSIC and implement it in an end-to-end network?
3) In the following hierarchical spectral–spatial feature learning, do features from different spatial locations and different channels contribute equally to the classification task? If no, how to data-driven emphasize spatial parts and channel parts that are useful for classifying a pixel and suppress parts that are useless, and implement them in an end-to-end network?

To deal with these problems, inspired by the visual attention mechanism of human beings, we proposed an end-to-end residual spectral–spatial attention network (RSSAN) for HSIC. As we know, human beings can allocate their limited resource to focus on the most important, informative, or salient part. In terms of spectral features, we design a spectral attention module (SeAM) to adaptively learn the weight of each spectral band for different raw input data, which implicitly applies feature selection to spectral vectors. In terms of spatial features, we want surrounding pixels having the same labels as the center pixel to have a great contribution to classify the center pixel while those having the different labels to contribute less. Thus, we design a spatial attention module (SaAM) to

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adaptively learn the importance of each surrounding pixel on classifying the center pixel, which enhances the contribution of surrounding pixels with the same labels as the center pixel on classifying the center pixel while weakens the contribution of those with different labels. In terms of discriminative feature learning, two attention modules are arranged sequentially and embedded into a CNN to emphasize meaningful features and suppress unnecessary ones. To alleviate the challenge of small training samples, we reuse the CNN parameters for two attention modules. To accelerate the training of the proposed model and avoid overfitting, we also adopt residual block and batch normalization in RSSAN. We validate the effectiveness of our RSSAN on three HSI data sets. Experimental studies demonstrate that RSSAN can achieve superior classification results.

The main contributions of this article are as follows.

1) We propose a learnable SeAM that implicitly implements band selection of HSI data, which can be learned in an end-to-end manner. This module can generate a spectral weight vector that reflects the importance of different spectral bands. The modulated spectral bands by this spectral weight vector can effectively improve the performance of classification.
2) We proposed a learnable SaAM that adaptively selects spatial information from different pixels in the neighborhood, which also can be learned in an end-to-end manner. This module can generate a spatial weight matrix that represents the significance of neighborhood pixels. The recalibrated spatial information by this spatial weight matrix can also effectively improve the classification results.
3) We proposed a learnable spectral-spatial attention module (SSAM) that refines the learned spectral–spatial features by sequentially stacking an SeAM and an SaAM. This module enhances more informative features and weakens less informative features by weighting the learned features.
4) We experimentally demonstrate the effectiveness of the proposed attention modules and illustrate that the proposed RSSAN outperforms eight compared methods on three HSI data sets.

The remainder of this article is organized as follows. Section II illustrates the details of the proposed model. Experiment studies and discussion of results are shown in Section III. Finally, the conclusion and future research direction are given in Section IV.

II. Methodology

In this section, we first introduce the framework of the proposed method for HSIC. Second, we introduce a basic CNN for feature learning and classification. Then, two attention modules: a SeAM and a SaAM, and residual attention learning are illustrated in detail. Finally, the loss function of the proposed method and its optimization method are given.

A. Overview of the Proposed Model

Let the HSI data set be $\mathcal{H} \in R^{h\times w\times d}$, where h, w, and d represent the height and width of the spatial dimensions and the number of the spectral bands, respectively. Suppose that the data set $\mathcal{H}$ contains N labeled pixels $\boldsymbol{X} = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N\} \in \mathbb{R}^{1\times 1\times d}$ and their corresponding set of one-hot label vectors $\boldsymbol{Y} = \{\boldsymbol{y}_1, \boldsymbol{y}_2, \ldots, \boldsymbol{y}_N\} \in \mathbb{R}^{1\times 1\times K}$, where K is the number of classes. To make full use of the spectral and spatial information provided by HSI, neighboring cubes centered at pixels in $\boldsymbol{X}$ form a new set of 3-D patch cubes $\boldsymbol{Z} = \{z_1, z_2, \ldots, z_N\} \in \mathbb{R}^{\omega\times\omega\times d}$, where ω is the width of patch cubes. In this article, we use each patch cube z_i in $\boldsymbol{Z}$ as the input to the proposed model to classify its corresponding center pixel $\boldsymbol{x}_i$ in HSI.

After the notation of HSI data, first, all available labeled data are randomly divided into *training*, *validation* and *testing* data sets denoted by $\boldsymbol{Z}_{\text{train}}$, $\boldsymbol{Z}_{\text{val}}$, and $\boldsymbol{Z}_{\text{test}}$, respectively, and corresponding label sets $\boldsymbol{Y}_{\text{train}}$, $\boldsymbol{Y}_{\text{val}}$, and $\boldsymbol{Y}_{\text{test}}$, respectively. Then, for each configuration of the proposed model and its hyperparameters, we only use $\boldsymbol{Z}_{\text{train}}$ to optimize the parameters of proposed model and test the trained model on $\boldsymbol{X}_{\text{val}}$ to find the best-trained model through cross-validation. Finally, the best-trained model is used to obtain three evaluation metrics of performance by $\boldsymbol{Z}_{\text{test}}$ and classify all pixels to form a classification map.

Fig. 1 shows the framework of the proposed model. To make full use of spectral and spatial information simultaneously, we extract pixel-centric 3-D patches as the input of CNN. The first layer of the proposed model is a spectral–spatial attention (SSA) module (red dotted rectangle in Fig. 1), which learns to find important spectral and spatial parts. Being different from implementing the attention modules on intermediate features (e.g., remove the layer in the red dotted rectangle in Fig. 1) in CNN for natural image processing in computer vision (CV), we directly apply the designed attention modules to the original HSI patches to capture their spectral attention and spatial attention for extracting important spectral and spatial information, respectively. To keep the original spectral–spatial information, we use residual learning [52] on this SSA without excessive features operation. The following is a convolutional module including a 3×3 convolution, a batch normalization, and an ReLU [53] activation function to learn low-level features.

As illustrated in CV [54]–[56], the embedding of attention module in residual learning module can enhance the learned features for the ultimate task. Inspired by the abovementioned conclusion, we design a residual spatial–spectral attention (RSSA) module (detail in the next section) to learn important features. Meanwhile, this RSSA module can be stacked to any depth. In this article, we stack two RSSA modules. Finally, to reduce the number of trainable parameters, we use a global average pooling (GAP) [57] on the learned features from the last RSSA module to aggregate spatial features. The output is fed to a softmax classifier. We will illustrate the abovementioned parts in the following sections in detail.

B. CNN and Residual Network (ResNet)

CNNs play an important role in many CV tasks. They can keep spatial information and learn locally invariable spatial features and nonlinear features by convolution, pooling, and

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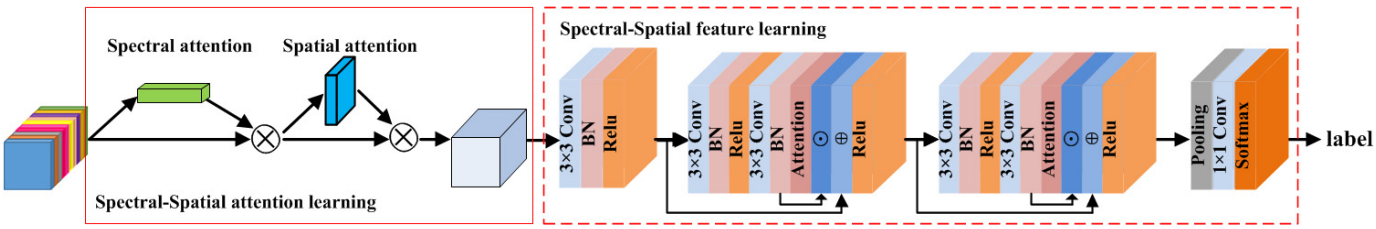


Fig. 1. Framework of the proposed RSSAN for HSIC of labeled pixels with a $7 \times 7 \times d$. d is the number of spectral bands of HSI. Given a raw 3-D patch, first, spectral–spatial attention learning (in red solid line rectangle) containing an SeAM and an SaAM aims to implicitly implement spectral band selection and adaptive selection of spatial information. Then, subsequent spectral–spatial feature learning (in the red dotted rectangle) progressively extracts hierarchical features for HSIC.

activation function, respectively. Using a convolution operation can largely reduce the number of trainable parameters compared with a fully connected (FC) feed-forward neural network [also called multilayer perceptron (MLP)]. CNNs are widely used in many image-processing problems, such as image classification [52], object detection, image segmentation, image super-resolution, and object tracking.

For a traditional CNN, a convolutional layer with an activation function can be formulated as follows:

$$\begin{aligned} \boldsymbol{h}_{i,j}^{l+1,k} &= f\left(\left(\boldsymbol{W}^{l+1,k} * \boldsymbol{h}^{l}\right)_{i,j} + \boldsymbol{b}^{l+1,k}\right) \\ &= f\left(\sum_{m=1}^{s_{l+1}} \sum_{n=1}^{s_{l+1}} \sum_{p=1}^{k_l} \boldsymbol{W}_{m,n,p}^{l+1,k} \cdot \boldsymbol{h}_{i+m-1,j+n-1,p}^{l} + \boldsymbol{b}^{l+1,k}\right) \end{aligned} \tag{1}$$

where $\boldsymbol{h}_{i,j,p}^{l}$ is the output feature map p at pixel (i, j) of the lth layer, and $\boldsymbol{W}^{l+1,k}$ and $\boldsymbol{b}^{l+1,k}$ are the kth convolutional kernel and bias of the $l+1$th layer, respectively. s_{l+1} is the size of convolutional kernel of the $l+1$th layer, and k_l is the number of feature maps of the lth layer. $f(\cdot)$ is an activation function, such as sigmoid function (Sigm), hyperbolic tangent function (Tanh), and rectified linear function (ReLU) [53]. In this article, we use ReLU as the activation function.

ResNet [52] is one of the famous and widely used CNNs. The residual module is easy to understand and optimize, which can be stack to any depth and embedded in any exiting CNN. The ResNet has been adopted as the baseline by many newly proposed modules in CNN, such SEnet [53] and CBAM [56]. In this article, we adopt the ResNet as the baseline. The residual module can be formulated as follows:

$$\boldsymbol{h}^{l+2} = \mathcal{F}\left(\boldsymbol{h}^{l}\right) + \boldsymbol{h}^{l} \tag{2}$$

where $\boldsymbol{h}^{l}$ and $\boldsymbol{h}^{l+2}$ represent the output of the lth and $l+2$th layers, respectively, and $\mathcal{F}$ can be one of or a series of operations, including convolution, pooling, and activation.

For all convolutional layers, we adopt a 3×3 convolution kernel with stride of 1 and pad of 1 without using any pooling, following by a batch normalization [58] and ReLU [53] activation function.

The input of RSSAN is a $7 \times 7 \times d$ patch, where d is the number of spectral bands. The first layer is 3×3 convolutions. Then, we use a stack of $6n$ layers with 3×3 convolutions on the feature maps of sizes $\{7, 7, 7\}$, respectively, with $2n$ layers for each feature map size. The numbers of filters are $\{16, 16, 16\}$, respectively. The network ends with a global average pooling, a k-d FC layer, and a softmax function, where k is the number of classes. In this article, we set n to 1 to avoid overfitting as only a little of training samples are used to train the network.

C. Spectral–Spatial Attention

Attention mechanism has been widely used in natural language processing (NLP) and CV, such as machine translation [59], image captioning [60], and image classification [54]–[56]. This mechanism can be implemented by two approaches: hard attention or soft attention. Given international features, hard attention aims to select useful parts in these features, while soft attention learns a weighting vector to weight all these features. Taking image classification problem using soft attention as an example, Wang *et al.* [54] used a scale attention to weight the output of convolutions with different filter sizes. A lightweight channel attention was proposed by squeezing and exciting channel features [55]. Moreover, spatial attention and channel attention was designed separately and cascaded [56]. In this article, we adopt these above two attention modules for the proposed model. In HSIC, there are some recently published works that incorporate visual attention with CNN. Haut *et al.* [61] introduced visual attention mechanism into HSIC in the first time, which use a bottleneck ResNet as a mask branch to calculate the mask and then multiply the mask to the learned features produced by a trunk branch composing by several residual blocks to extract useful information for HSIC. Mou and Zhu [62] designed an SeAM to adaptively recalibrate spectral bands by enhancing informative bands and weakening useless bands. Fang *et al.* [63] incorporated a spectral-wise attention module with a 3-D dense convolutional network to enhance the distinguishability of spectral features. In this article, we consider not only adaptively recalibrating spectral bands but also the adaptive selection of spatial information and the refinement of learned spectral–spatial features.

Given a feature map $\boldsymbol{h}^{l} \in \mathbb{R}^{H \times W \times C}$ (for convenience, we will ignore the superscript l), we generate a 1-D spectral attention map $\boldsymbol{M}_{se} \in \mathbb{R}^{1 \times 1 \times C}$ and a 2-D spatial attention map $\boldsymbol{M}_{sa} \in \mathbb{R}^{H \times W \times 1}$, as shown in Fig. 2, where H, W, and C represent height, width, and channel, respectively. The abovementioned two attention modules correspond to band selection and spatial selection in HSI, respectively. The overall

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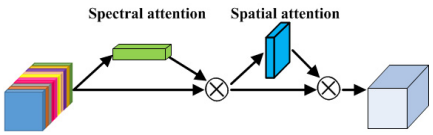


Fig. 2. Spectral–spatial attention. Two attention modules are arranged sequentially. Given input or intermediate features, spectral attention is computed and multiplied to them as the input of the following spatial attention. Similar operation as spectral attention is done as the output of SSA module.

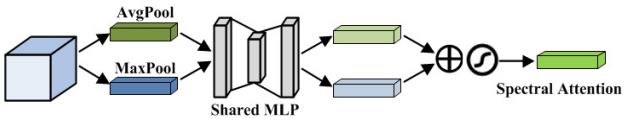


Fig. 3. Spectral attention module. This module utilizes both max-pooling outputs and average-pooling outputs with a two-layer weight-shared bottleneck network.

attention computation can be summarized as

$$\begin{aligned} \boldsymbol{h}' &= \boldsymbol{M}_{se}(\boldsymbol{h}) \otimes \boldsymbol{h} \\ \boldsymbol{h}'' &= \boldsymbol{M}_{sa}(\boldsymbol{h}') \otimes \boldsymbol{h}' \end{aligned} \tag{3}$$

where $\otimes$ denotes the elementwise multiplication. Detail of attention computation is given in the following section.

D. Spectral Attention Module

The SeAM aims to emphasize those spectral bands that help to feature representation and final classification. In order to achieve the above purpose, we need to design a data-driven function mapping input features to a spectral weight vector, which implies the weight or importance of each spectral band. Meanwhile, the mapping function should consider the interaction between different spectral bands. The SeAM is shown in Fig. 3. To generate spectral-wise summary statistics $\boldsymbol{z}^{\text{avg}} \in \mathbb{R}^{1\times 1\times C}$, the global average pooling is operated on each spectral feature across spatial dimensions $H \times W$, as done in [55]. The cth element of $\boldsymbol{z}^{\text{avg}}$ is calculated by

$$z_c^{\text{avg}} = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} \boldsymbol{h}_c(i, j) \tag{4}$$

where $\boldsymbol{h}_c(i, j)$ is the value at position (i, j) of the cth channel $\boldsymbol{h}_c$. In addition, CBAM [56] states that the global max pooling also has complementary global information for global average pooling. We also adopt the global max pooling in this article. Same as the global average pooling, the cth channel of $\boldsymbol{z}^{\max}$ is calculated by

$$z_c^{\max} = \max(\boldsymbol{h}_c). \tag{5}$$

To fully capture the interaction between spectral bands, limit model complexity, and aid generalization, a bottleneck with two FC layers around the nonlinearity is operated on $\boldsymbol{z}$. The first FC layer is a dimensionality-reduction layer parameterized by W_1 with reduction ratio r and a ReLU.

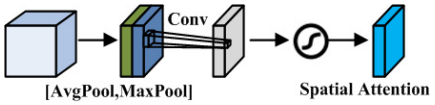


Fig. 4. Spatial attention module. It utilizes both max-pooling outputs and average-pooling outputs that are pooled along the spectral axis and stacked horizontally them as the input of a convolution layer.

The second is a dimensionality-increasing layer parameterized by W_2

$$\boldsymbol{s} = \sigma(g(\boldsymbol{z}, \boldsymbol{W})) = \sigma(\boldsymbol{W}_2(\delta(\boldsymbol{W}_1 \boldsymbol{z}))) \tag{6}$$

where δ and σ refer to the ReLU [53] and sigmoid function, respectively, $W_1 \in \mathbb{R}^{(C/r)\times C}$, and $W_2 \in \mathbb{R}^{C\times(C/r)}$, reduction ratio r set to 8 in the experiment.

To reduce the number of parameters, we shared the parameters $\{W_1, W_2\}$ of two FC layers for the output of both global average pooling and global max pooling. Then, the outputs of two FC layers $\boldsymbol{s}_c^{\text{avg}}$ and $\boldsymbol{s}_c^{\max}$ are added together

$$\boldsymbol{s} = \boldsymbol{s}^{\text{avg}} + \boldsymbol{s}^{\max}. \tag{7}$$

The final output of the SeAM is obtained by rescaling the input features with the output activation s

$$\boldsymbol{h}'_c = \boldsymbol{F}_{\text{scale}}(\boldsymbol{h}_c, s_c) = s_c \cdot \boldsymbol{h}_c \tag{8}$$

where $\boldsymbol{h}' = \boldsymbol{h}'_1, \boldsymbol{h}'_2, \ldots, \boldsymbol{h}'_C$, and $\boldsymbol{F}_{\text{scale}}(\boldsymbol{h}_c, s_c)$ refers to spectral-wise multiplication between the feature map $\boldsymbol{h}_c \in \mathbb{R}^{H\times W\times 1}$ and the scalar s_c.

E. Spatial Attention Module

The SaAM aims to enhance spatial information from the pixels in the neighborhood with the same class label as the center pixel and suppress those with a different class label. Thus, the ideal output of the SaAM should be a matrix of the same height and weight as the input patch where the value equal to 1 if the pixel in this location with the same label as the center, otherwise 0.

Fig. 4 shows the SaAM. As in CBAM [56], we first apply the global max pooling and global average pooling on the input across spectral channels

$$\boldsymbol{sa}_{i,j}^{\text{avg}} = \frac{1}{C} \sum_{c=1}^{C} \boldsymbol{h}'_c(i, j) \tag{9}$$

and

$$\boldsymbol{sa}_{i,j}^{\max} = \max(\boldsymbol{h}'_c) \tag{10}$$

where $\boldsymbol{h}'_c(i, j)$ is the value at position (i, j) of the cth channel $\boldsymbol{h}'_c$. Then, the two outputs are concatenated horizontally as the input of a new convolutional layer followed by a sigmoid activation function

$$\boldsymbol{sa} = \text{sigm}([\boldsymbol{sa}^{\text{avg}}, \boldsymbol{sa}^{\max}] * \boldsymbol{W}_3) \tag{11}$$

where $*$ is the convolution operation.

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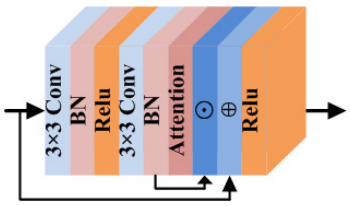


Fig. 5. Residual spatial–spectral attention (RSSA) module. Attention means a SSA in Fig. 2.

The final output of the SaAM is obtained by rescaling the input features with the output activation $\boldsymbol{sa}$

$$\boldsymbol{h}'' = \boldsymbol{h}' \star \boldsymbol{sa} \tag{12}$$

where $\star$ is the spatialwise multiplication operation between the feature map $\boldsymbol{h}'_{i,j} \in \mathbb{R}^{1\times 1\times C}$ and the scalar $\boldsymbol{sa}_{i,j}$.

F. Residual Spatial–Spectral Attention Module

In view of the superior performance of the residual block in ResNet [52], the SSA block is embedded into the residual block, as shown in Fig. 5.

The residual SSA module consists of two convolutional layers, two batch normalizations [58], a SSA block, a skip connection, and two ReLU activation functions. The arrangement of these basic layers or blocks is shown in Fig. 5 and formulated as follows:

$$\boldsymbol{h}^{p+2} = \text{Relu}\left(\boldsymbol{h}^{p} + \boldsymbol{h}_{\text{SSA}}^{p+2'} \cdot \boldsymbol{h}^{p+2'}\right) \tag{13}$$

$$\boldsymbol{h}_{\text{SSA}}^{p+2'} = F_{\text{SSA}}\left(\boldsymbol{h}^{p+2'}; \boldsymbol{\theta}_{\text{SSA}}\right) \tag{14}$$

$$\boldsymbol{h}^{p+2'} = \text{BN}\left(\boldsymbol{h}^{p+1} * \boldsymbol{W}^{p+2} + \boldsymbol{b}^{p+2}\right) \tag{15}$$

$$\boldsymbol{h}^{p+1} = \text{Relu}\left(\text{BN}\left(\boldsymbol{h}^{p} * \boldsymbol{W}^{p+1} + \boldsymbol{b}^{p+1}\right)\right) \tag{16}$$

where $\boldsymbol{\theta}_{\text{SSA}} = \{\boldsymbol{W}_{SeAM}^{p+2}, \boldsymbol{b}_{SeAM}^{p+2}, \boldsymbol{W}_{SaAM}^{p+2}, \boldsymbol{b}_{SaAM}^{p+2}\}$, $\boldsymbol{\theta} = \{\boldsymbol{W}^{p+1}, \boldsymbol{b}^{p+1}, \boldsymbol{W}^{p+2}, \boldsymbol{b}^{p+2}\}$, $\boldsymbol{h}^{p}$ represents the output features of the pth layer, and $\boldsymbol{W}^{p+1}$ and $\boldsymbol{b}^{p+1}$ denote the convolutional kernels and bias in the $(p+1)$th layer, respectively. Relu$(\cdot)$, $BN(\cdot)$, and $F_{\text{SSA}}(\cdot)$ represent the ReLU activation function, the batch normalization, and the SSA operation, respectively.

G. Loss Function and Optimization

To optimize the above-given model, a suitable objective function is needed. Cross entropy is common used loss function for the classification problems. Here, we also use the cross entropy as loss function of RSSAN for classification of HSI. The loss function for cross entropy is given as follows:

$$\text{Loss}_{\text{CE}} = -\frac{1}{M}\sum_{m=1}^{M}\sum_{c=1}^{C} \boldsymbol{y}_c^m \log\left(\hat{\boldsymbol{y}}_c^m\right) \tag{17}$$

where $\boldsymbol{y}$ and $\hat{\boldsymbol{y}}$ are the truth and predict labels, respectively, C is the number of classes, and M is the number of samples in a minibatch. The parameters in the model are updated by backpropagation and stochastic gradient descent.

III. Experiments

In this section, we first describe three HSI data sets, then analyze four factors that influence the performance of the proposed model, compare the proposed model with the state-of-the-arts based on deep learning methods, analyze the influence of attention module on the proposed model, and, finally, discuss the comparison result.

A. Data Set Description

In the experiments, we consider three publicly available HSIs[1]: Indian Pines (IN), University of Pavia (UP), and Kennedy Space Center (KSC), as listed in Table I.

1) Indian Pines (IN): This data set was gathered by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor over the Indian Pines test site in Northwestern Indiana on June 1992. It consists of 224 spectral bands with 145 × 145 pixels. It has a spectral resolution of 10 nm ranging from 0.4 to 2.5 μm and a spatial resolution of 20 m/pixel. The ground truth contains 16 vegetation classes with 10 249 labeled pixels. After removing 20 bands covering the region of water absorption and four zero-bands, the remaining 200 bands are used for classification in this article. Fig. 10 shows the false-color image and ground-truth map. Table II lists the numbers of training and test samples for each class.

2) University of Pavia (UP): This data set was acquired by the Reflective Optics System Imaging Spectrometer (ROSIS) over the urban area of the University of Pavia in Northern Italy on July 8, 2002. It consists of 115 spectral bands with 610 × 340 pixels. It has a spectral resolution of 4 nm ranging from 0.43 to 0.86 μm and a spatial resolution of 1.3 m/pixel. The ground truth contains nine urban land-cover types with 42 776 labeled pixels. After removing 12 noisy bands, the remaining 103 bands are used for classification in this article. Fig. 11 shows the false-color image and ground-truth map. Table III lists the numbers of training and test samples for each class.

3) Kennedy Space Center (KSC): This data set was acquired by the National Aeronautics and Space Administration (NASA) Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) instrument over the Kennedy Space Center (KSC), FL, USA, on March 23, 1996. It consists of 224 spectral bands with 512 × 614 pixels. It has a spectral resolution of 10 nm ranging from 0.4 to 2.5 μm and a spatial resolution of 18 m/pixel. The ground truth contains 13 upland and wetland classes with 5211 labeled pixels. After removing water absorption and low signal-to-noise ratio (SNR) bands, the remaining 176 bands are used for classification with 48 bands discarded in this article. Fig. 12 shows the false-color image and ground truth map. Table IV lists the numbers of training and test samples for each class.

Three data sets have different sizes of available labeled samples. There are 10 249 and 5211 labeled samples for the IN and KSC data sets, respectively, while 42 776 labeled samples for the UP data set. Compared with the first two, the latter has a relatively large data. We used different proportions of

[1]Data set is available on http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes

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TABLE I
LIST OF THREE DATA SETS

Data	Spatial	Spctral	Class	Samples		Resolution		Wavelength	Sensor	Date
				Labeled	Unlabeled	Spectral	Spatial			
IN	145 × 145	200	16	10,249	10,776	10 nm	20 m	400-2500 nm	AVIRIS	1-Jun-92
UP	610 × 340	103	9	42,776	164,624	3.74 nm	1.3 m	430-860 nm	ROSIS	8-Jul-02
KSC	512 × 614	176	13	5,211	309,157	10 nm	20 m	400-2500 nm	AVIRIS	23-Mar-96

TABLE II
CLASS NAME AND SAMPLE NUMBER OF EACH CLASS FOR IN DATA SET

No.	Class Name	Train	Val	Test
1	Alfalfa	10	4	32
2	Corn-notill	286	133	1,009
3	Corn-mintill	166	99	565
4	Corn	48	21	168
5	Grass-pasture	97	52	334
6	Grass-trees	146	73	511
7	Grass-pasture-mowed	6	3	19
8	Hay-windrowed	96	48	334
9	Oats	4	1	15
10	Soybean-notill	195	93	684
11	Soybean-mintill	491	242	1,722
12	Soybean-clean	119	56	418
13	Wheat	41	24	140
14	Woods	253	123	889
15	Buildings-Grass-Trees-Drives	78	41	267
16	Stone-Steel-Towers	19	12	62
	Total	2,055	1,025	7,169

TABLE III
CLASS NAME AND SAMPLE NUMBER OF EACH CLASS FOR UP DATA SET

No.	Class Name	Train	Val	Test
1	Asphalt	664	681	5,286
2	Meadows	1,865	1,858	14,926
3	Gravel	210	218	1,671
4	Trees	307	311	2,446
5	Painted metal sheets	135	135	1,075
6	Bare Soil	503	489	4,037
7	Bitumen	133	147	1,050
8	Self-Blocking Bricks	369	353	2,960
9	Shadows	95	89	763
	Total	4,281	4,281	34,214

label samples as training samples for each data set to test the performance of the proposed model using different sizes of training samples. As in work [8], we randomly select 20%, 10%, and 70% of the whole labeled samples as training, validation, and testing sets for the IN and KSC data sets while 10%, 10%, and 80% for the UP data set. In addition, all the input data of three HSI data sets are standardized to zero mean value with unit variance. The details of the number of each class for training, validation, and testing sets are listed in Tables II–IV.

TABLE IV
CLASS NAME AND SAMPLE NUMBER OF EACH CLASS FOR KSC DATA SET

No.	Class Name	Train	Val	Test
1	Scrub	153	80	528
2	Willow swamp	49	24	170
3	CP hammock	52	27	177
4	Slash pine	51	28	173
5	Oak/Broadleaf	33	14	114
6	Hardwood	46	22	161
7	Swamp	21	11	73
8	Graminoid marsh	87	42	302
9	Spartina marsh	104	64	352
10	Cattail marsh	81	42	281
11	Salt marsh	84	32	303
12	Mud flats	101	49	353
13	Water	186	86	655
	Total	1,048	521	3,642

B. Experimental Configuration

In order to evaluate the performance of the proposed RSSAN model, the experiments are implemented on a computer with a Intel Xeon CPU E5-2630 v3 at 2.40 GHz × 32 with 64-GB RAM and a NVIDIA GeForce GTX TITAN X graphical processing unit (GPU) with 12-GB RAM. The software environment is the system of Ubuntu 14.04 ultimate 64 bit and deep learning frameworks of TensorFlow [64] and Keras [65].

Optimization is performed by RMSProp optimizer [66] with the batch size of 16, and early stopping is used to avoid overfitting. We used the He normal distribution initialization method [67] to initialize the weights and bias for all CNNs.

To quantitatively evaluate the proposed method, we use class-specific accuracy, overall accuracy (OA), average accuracy (AA), and kappa coefficient (κ) as the evaluation metrics.

C. Parameter Setting

We analyze some factors that influence the training progress and classification performance of the trained RSSAN. These

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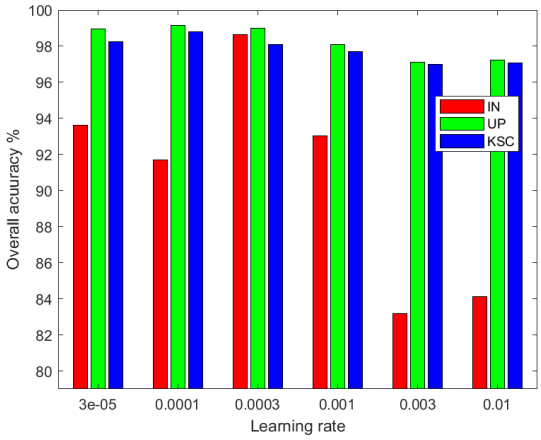


Fig. 6. OA (%) of RSSANs with different learning rates in the IN, UP, and KSC data sets.

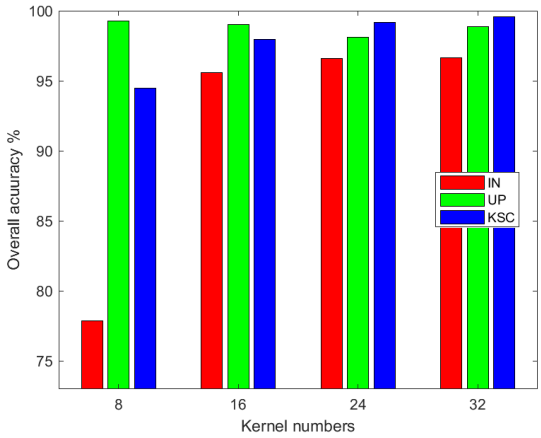


Fig. 7. OA (%) of RSSANs with different kernel numbers of convolutional filters in the IN, UP, and KSC data sets.

factors are the learning rate, the kernel number of convolutional layers, the spatial input size, and the training samples proportion, which were also named hyperparameters. For each combination of the abovementioned hyperparameters, the models with the highest classification performance on the validation data set were preserved and used for the following comparison. Considering the small size of training samples, we use a small batch size of 16. We run each experiment 200 epochs in the following experiments.

1) Learning Rates: The learning rate controls the step of gradient descent in training progress. It plays a great role in the convergent speed and convergence of the training progress. We use the grid search to find the best learning rate for the proposed model on each data set. Here, we consider the learning rate sets {0.01, 0.003, 0.001, 0.0003, 0.0001, 0.00003}. The results are shown in Fig. 6. Based on the abovementioned results, the learning rate is set to 0.0003 for IN and UP data set and 0.0001 for the KSC data set in the following experiments.

2) Kernel Numbers of Convolutional Filters: The kernel numbers of convolution filters decide the representation capacity and computational consumption of RSSAN. We use the same kernel number in each convolutional layer for RSSAN. Here, we consider different kernel numbers {8, 16, 24, 32} to find an optimum kernel number for each convolutional layer. As shown in Fig. 7, the bigger the kernel number is, the better OA is for the IN and KSC data sets, while vice versa for the UP data set. The models with 32 kernels achieved the best performance on the IN and KSC data sets, and the models with eight kernels obtained the highest classification accuracy on the UP data set.

3) Spatial Input Sizes: The spatial input size decides how much spatial information around one pixel is used to classify a pixel. To evaluate the influence of the spatial input size on the performance of RSSAN, we consider a large set of spatial input sizes {3, 5, 7, 9, 11, 13, 15, 17, 21}. As shown in Fig. 8, RSSANs perform robustly for different spatial input size if these sizes are equal to or larger than 9×9 on the UP and KSC data sets.

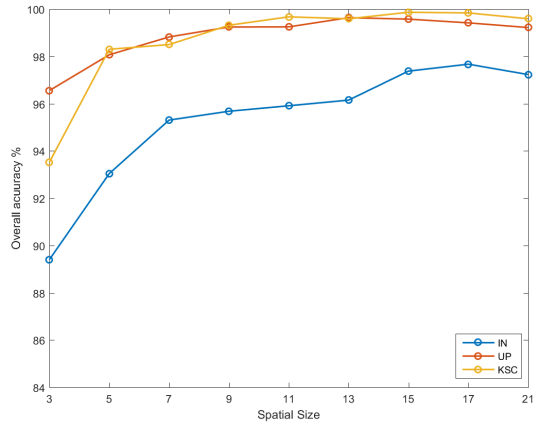


Fig. 8. OA (%) of RSSANs with different spatial input sizes in the IN, UP, and KSC data sets.

4) Training Sample Proportions: We consider using 1%, 3%, 5%, 10%, 20%, and 30% of labeled pixels as the training set to train the RSSAN, respectively. As shown in Fig. 9, the OA increases as the training samples proportion increases for all three HSI data sets. Compared with the other eight methods, the RSSAN performs the best on all training sample proportions and three data sets.

D. Comparison With the State-of-the-Arts

In this section, we compared the proposed RSSAN model with two traditional methods: SVM [68] and extended multi-APs (EMAP) [69], and six deep learning-based methods: SAE-LR[2] [9], CNN[3] [7], a deep feature fusion network (DFFN) [70], a CNN with mapping layers (MCNN) [71], 3-D-CNN[4] [72], and SSRN[5] [8]. SAE-LR was the first work that applies the deep learning to HSIC and was implemented

[2]SAE-LR Theano code: https://github.com/hantek/deeplearn_hsi

[3]CNN Keras+TensorFlow ipython code: https://github.com/KonstantinosF/Classification-of-Hyperspectral-Image

[4]3-D-CNN Keras+Theano code: https://github.com/YellowManb/CNN_HSIC

[5]SSRN Keras+TensorFlow code: https://github.com/zilongzhong/SSRN

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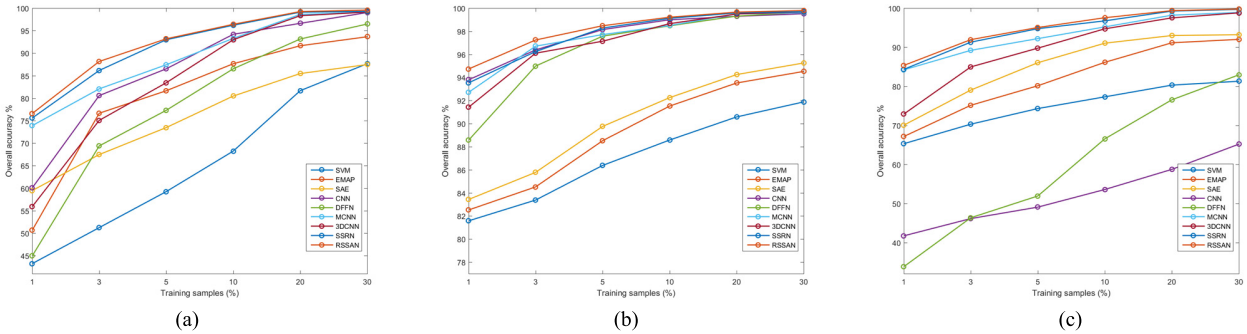


Fig. 9. OA (%) of RSSANs with different training sample proportions in the IN, UP, and KSC data sets.

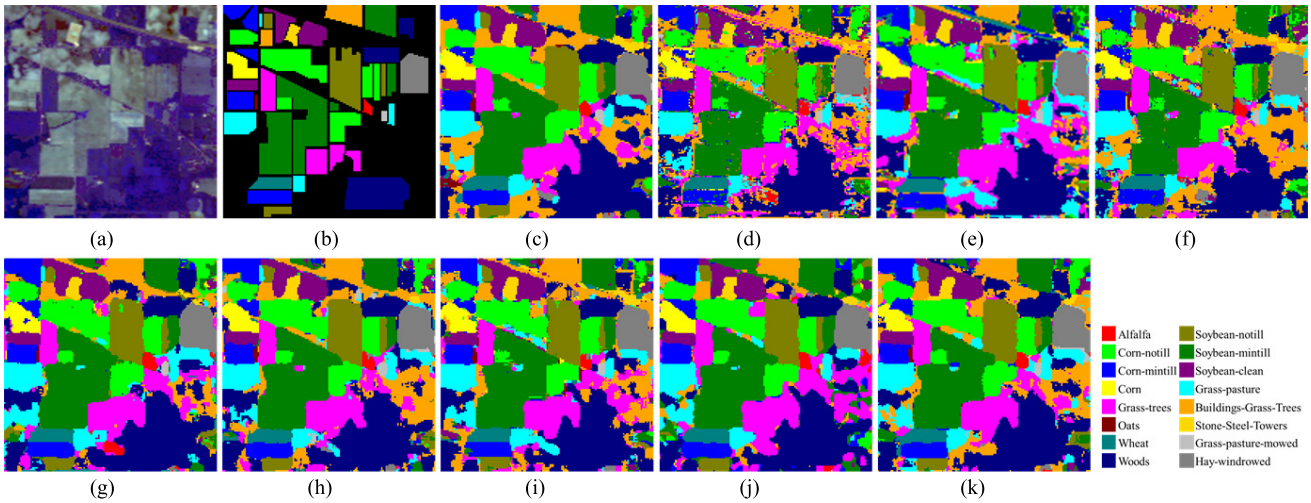


Fig. 10. Classification maps for the IN data set. (a) False-color image (12, 25, 36). (b) Ground-truth map. (c)–(k) Classification maps of SVM, EMAP, SAE, CNN, DFFN, MCNN, 3-D-CNN, SSRN, and RSSAN.

with Theano [73]. The latter five methods are based on 2-D-CNN or 3-D-CNN and implemented with Keras [65] using TensorFlow [64] as a back end. The 3-D-CNN [72] and SSRN [8] all uses the 3-D-CNN as baseline.

1) Quantitative Evaluation: The quantitative metric comparisons of different methods are shown in Tables V–VII. Compared with two traditional methods, the deep learning-based methods achieve better performance on three data set, except that EMAP outperforms SAE-LR on the IN data set. This is due to the fact that deep learning combines hierarchical feature learning with classifier learning, so the deep learning-based methods, such as CNN, can learn more discriminative and abstract high-level features that are useful for HSIC, while SAE only uses unlabeled data to implement unsupervised feature learning, separating from subsequent classification tasks, so it, sometimes, gets weaken classification result compared with EMAP. The proposed RSSAN achieves state-of-the-art results on all three data sets. Especially, the proposed 2-D-CNN-based RSSAN outperforms the 3-D-CNN-based model SSRN on three data sets. As we know, spectral feature is important for the classification of HSI, and 3-D-CNN can fully learn the spectral feature compared with the traditional 2-D-CNN. However, the proposed model uses an SeAM to strengthen these useful spectral bands and weaken those useless or harmless spectral bands, so RSSAN achieved better accuracy than SSRN.

2) Qualitative Evaluation: Figs. 10–12 visualize the false-color images of the original HSI, their corresponding ground-truth maps, and the best classification results of different methods on three data sets. On three data sets, the classification maps of SAE, EMAP, CNN, DFFN, MCNN, and 3-D-CNN have some dot noises in some classes although these methods simultaneously consider the spectral and spatial feature, while SSRN generates more smooth classification maps than these methods as it considers the spectral and spatial feature learnings consecutively. Compared with other methods, the RSSAN generated the most accurate and smooth classification maps, especially in the boundary of two different classes, because the RSSAN uses spatial attention to learn the relationship between the center pixel and surrounding neighboring pixels.

E. Ablation Study

In this section, we study the role of an attention module in the RSSAN model.

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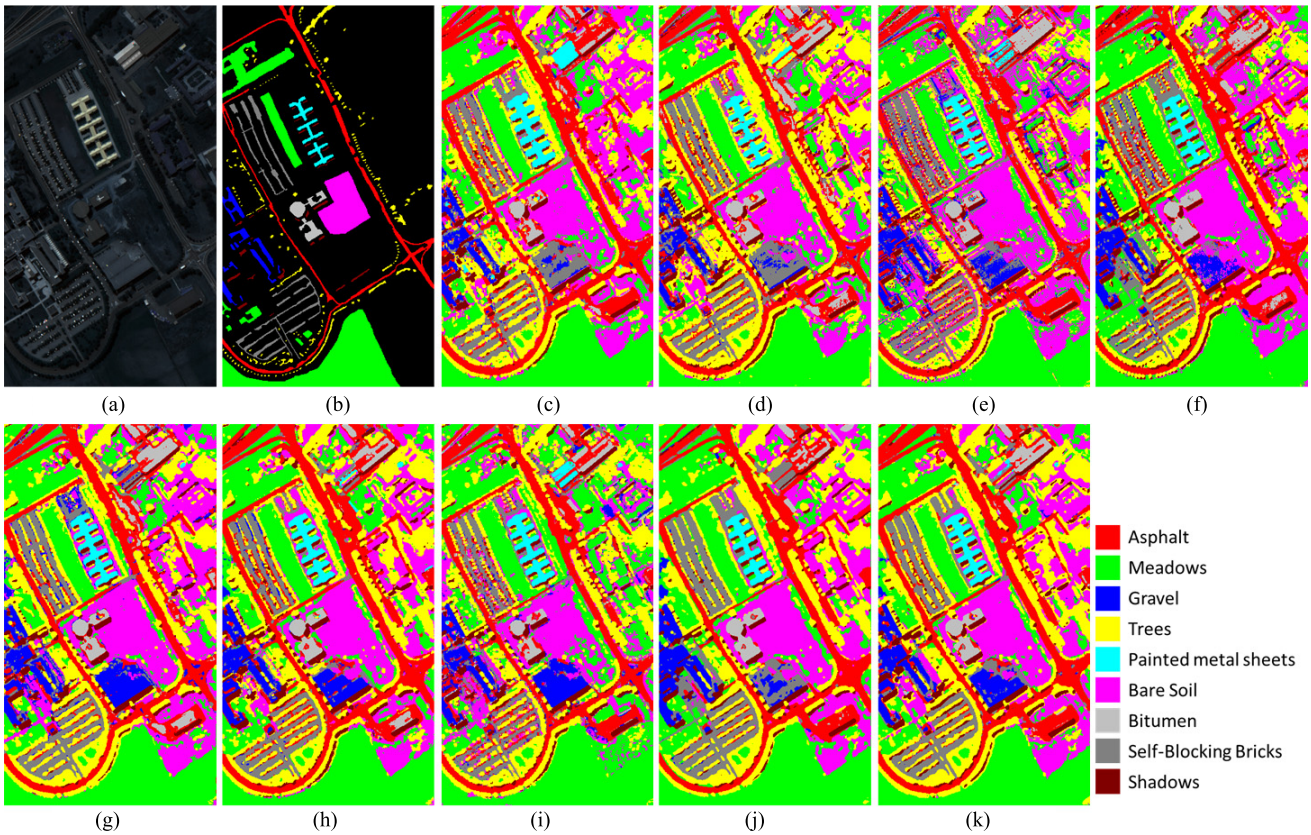


Fig. 11. Classification maps for the UP data set. (a) False-color image (12, 25, 36). (b) Ground-truth map. (c)–(k) Classification maps of SVM, EMAP, SAE, CNN, DFFN, MCNN, 3-D-CNN, SSRN, and RSSAN.

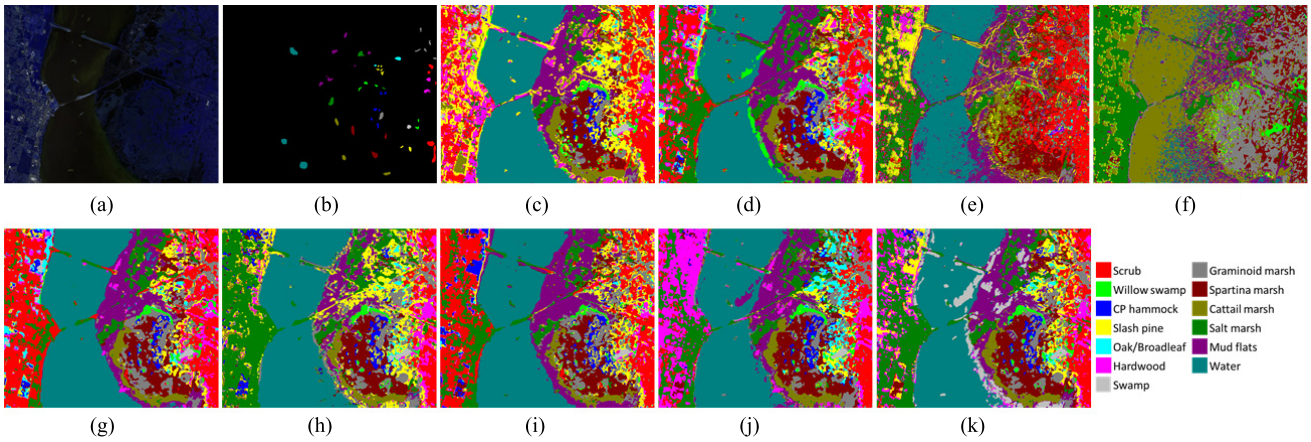


Fig. 12. Classification maps for the KSC data set. (a) False-color image (12, 25, 36). (b) Ground-truth map. (c)–(k) Classification maps of SVM, EMAP, SAE, CNN, DFFN, MCNN, 3-D-CNN, SSRN, and RSSAN.

1) First Attention Layer: We compare the RSSAN with or without an attention module as the first layer. As shown in Table VIII, the RSSAN with the first attention layer (see red solid line rectangle in Fig. 1) surpasses the RSSAN without first attention layer in terms of classification accuracy on all three HSI data sets. These results demonstrate that the first attention layer can select the useful original HSI data for following feature learning.

2) Attention Module: We compare the RSSAN with the RSSAN without spatial–spectral attention module (RSSAN-None), RSSAN only with spectral attention module (RSSAN-SE), RSSAN only with spatial attention module (RSSAN-SA), and RSSAN with spectral–spatial attention module (RSSAN-SASE) that first implements spatial attention then spectral attention. The results are shown in Table IX. For the UP data set, we found that RSSAN with spatial attention gets better OA than spectral attention; as this data set has a high spatial resolution of 1.3 m/pixel, neighboring pixels have a great influence on the classification of the center pixel.

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TABLE V
CLASSIFICATION RESULTS OF DIFFERENT METHODS FOR LABELED PIXELS OF THE IN DATA SET

Class	SVM	EMAP	SAE-LR	CNN	DFFN	MCNN	3D-CNN	SSRN	**RSSAN**
OA(%)	81.67±0.65	91.65±0.63	85.47±0.58	97.41±0.43	98.52±0.23	98.30±0.20	97.56±0.43	99.19±0.26	**99.46±0.04**
AA(%)	79.84±3.37	95.15±1.05	86.34±1.14	97.39±0.56	98.32±0.26	97.40±0.40	99.23±0.19	98.93±0.59	**99.30±0.26**
$\kappa\times$ 100	78.76±0.77	90.46±0.68	83.42±0.66	97.05±0.49	97.69±0.74	98.00±0.30	97.02±0.52	99.07±0.30	**99.38±0.05**
1	96.78	94.87	81.82	**100.00**	97.56	98.30	100	97.82	99.38
2	78.74	84.10	82.16	97.27	97.75	96.20	96.34	99.17	**99.37**
3	82.26	95.66	77.54	98.00	98.31	95.30	99.49	**99.53**	99.25
4	99.03	98.37	68.11	92.81	98.12	94.20	**100**	97.79	98.50
5	93.75	95.53	94.60	99.25	98.66	98.20	**99.91**	99.24	99.68
6	85.96	96.84	94.45	99.52	99.62	98.70	99.75	99.51	**99.86**
7	40.00	**100.00**	94.70	97.58	97.60	96.40	**100**	98.70	99.55
8	91.80	99.32	94.36	99.00	**100.00**	**100.00**	**100**	99.85	**100.00**
9	0.00	**100.00**	82.56	96.95	93.89	92.20	**100**	98.50	97.46
10	86.00	90.63	81.28	95.38	98.20	97.80	98.72	98.74	**99.11**
11	70.94	89.16	84.47	97.72	98.76	**99.60**	95.52	99.30	99.34
12	74.73	86.88	83.77	97.13	96.00	97.20	**99.47**	98.43	98.55
13	99.04	98.77	96.42	99.65	99.51	**100.00**	**100**	**100.00**	99.57
14	94.29	94.13	92.27	97.95	**99.82**	99.70	99.55	99.31	99.39
15	85.11	98.18	80.63	92.30	98.65	98.80	**99.54**	99.20	97.87
16	96.78	**100.00**	81.82	**100.00**	90.60	95.80	99.34	97.82	98.44

TABLE VI
CLASSIFICATION RESULTS OF DIFFERENT METHODS FOR LABELED PIXELS OF THE UP DATA SET

Class	SVM	EMAP	SAE-LR	CNN	DFFN	MCNN	3D-CNN	SSRN	**RSSAN**
OA(%)	90.58±0.47	93.52±0.34	94.25±0.18	98.85±0.15	98.73±0.31	99.50±0.20	99.54±0.11	99.79±0.09	**99.89±0.03**
AA(%)	92.99±0.36	94.82±0.40	93.34±0.39	98.40±0.30	98.31±0.41	99.30±0.10	99.66±0.11	99.66±0.17	**99.81±0.07**
$\kappa\times$ 100	87.21±0.70	91.65±0.49	92.35±0.25	98.47±0.20	97.24±0.53	99.30±0.20	99.41±0.15	99.72±0.12	**99.85±0.04**
1	87.24	91.30	94.59	98.98	98.82	99.60	99.36	**99.92**	99.77
2	89.93	91.83	96.44	99.45	99.74	99.40	99.36	**99.96**	99.91
3	86.48	72.22	84.57	96.04	98.14	99.40	**99.69**	98.46	99.36
4	99.95	99.80	97.37	99.58	94.43	99.80	99.63	99.69	**99.87**
5	95.78	99.93	99.60	99.39	99.14	99.90	99.95	**99.99**	99.92
6	97.69	99.62	99.39	99.70	99.69	99.80	**99.96**	99.94	**99.96**
7	95.44	99.70	88.57	97.18	96.68	98.80	**100**	99.82	99.59
8	84.40	99.27	85.66	95.73	98.95	98.90	**99.65**	99.22	98.41
9	**100.00**	99.79	99.88	99.56	89.54	98.40	99.38	99.95	99.99

F. Discussion

First, we introduce an attention mechanism-based deep neural network into HSIC. The experiment demonstrates that the RSSAN using the 2-D-CNN-based framework with attention module can achieve superior results compared with 3-D-CNN-based frameworks, such as 3-D-CNN and SSRN methods.

Second, we introduce residual learning block and batch normalization after each convolution, without using dropout in the whole network. The experiment illustrates that even in a limited small number of training samples, the RSSAN also avoids overfitting and achieves almost the state-of-the-art classification accuracy.

Finally, ablation study on attention module shows that for some HSI data sets that have more redundancy spectral bands and lower spatial resolution, spectral features are obviously superior to spatial features (e.g., the IN data set), while for the others HSI data sets that have less redundancy spectral bands and higher spatial resolution, and vice versa (e.g., the UP and KSC data sets). However, an appropriate arrangement or a combination of spatial features and spectral features may achieve better than both of them.

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TABLE VII
CLASSIFICATION RESULTS OF DIFFERENT METHODS FOR LABELED PIXELS OF THE KSC DATA SET

Class	SVM	SAE-LR	EMAP	CNN	DFFN	MCNN	3D-CNN	SSRN	**RSSAN**
OA(%)	80.29±0.58	92.99±0.82	91.23±0.59	97.08±0.47	82.05±0.59	98.49±0.65	96.31±1.25	99.61±0.22	**99.74±0.12**
AA(%)	65.64±0.86	89.76±1.25	89.17±1.64	95.09±0.70	69.58±0.63	97.12±0.62	94.68±1.97	99.33±0.57	**99.56±0.19**
$\kappa\times$ 100	77.98±0.65	92.18±0.91	90.25±0.54	96.74±0.53	79.89±0.86	98.32±0.74	95.90±1.39	99.56±0.25	**99.71±0.13**
1	92.16	93.04	94.09	99.00	98.31	99.44	91.71	99.70	**99.83**
2	86.16	92.04	86.15	98.48	18.24	98.82	89.73	**99.88**	99.76
3	42.55	85.59	98.05	92.16	88.27	**100.00**	92.16	99.00	98.07
4	67.69	72.12	67.33	81.84	39.77	82.95	86.94	**98.26**	96.89
5	0.00	82.20	71.32	85.38	79.65	84.07	94.79	**99.03**	97.52
6	54.71	83.15	89.13	90.96	39.38	**100.00**	90.92	99.43	98.72
7	0.00	76.46	95.24	93.21	0.00	97.30	91.57	97.03	**98.71**
8	65.12	94.10	88.70	98.21	59.27	**100.00**	96.22	99.54	99.57
9	67.82	94.57	95.43	99.04	86.26	**100.00**	99.53	99.70	99.95
10	93.40	98.91	100.00	99.85	95.76	**100.00**	99.81	99.96	**100.00**
11	100.00	98.39	93.45	98.89	100.00	**100.00**	97.69	99.80	99.83
12	83.75	96.42	84.12	99.43	99.72	**100.00**	97.69	**100.00**	99.60
13	100.00	99.83	96.23	99.79	100.00	**100.00**	100	**100.00**	99.98

TABLE VIII
OA (%) OF RSSANS WITH/WITHOUT AN ATTENTION MODULES AS THE FIRST LAYER

First Layer	IN	UP	KSC
w Attention	94.94±8.50	99.11±0.53	98.21±0.87
w/o Attention	93.86±4.88	98.95±0.36	97.76±1.71

TABLE IX
OA (%) OF RSSANS WITH DIFFERENT ATTENTION MODULES

Attention Module	IN	UP	KSC
None	95.02±3.34	99.26±0.40	97.65±1.67
Spectral Attention	95.67±5.18	99.20±0.31	96.79±3.36
Spatial Attention	89.68±8.38	99.30±0.27	97.52±1.54
Spatial + Spectral	94.01±8.21	99.34±0.27	97.72±1.09
Spectral + Spatial	95.29±8.33	99.27±0.33	98.63±0.36

IV. CONCLUSION

In this article, we proposed a novel RSSAN deep learning framework for HSIC, which uses ResNet as a baseline to learn spectral and spatial features and uses soft attention mechanism to implicitly implement the selection and enhancement of spectral bands and spatial information. First, the original HSI patches are direct as the input of the proposed model, which is an end-to-end trainable 2-D CNN-based framework for HSIC. Second, spectral attention and spatial attention are designed to select useful spectral bands and strength "good" pixels and weaken "bad" pixels in the neighborhood for HSIC. Then, the spatial–spectral attention is embedded into the residual block to accelerate the training of the proposed model and reduce the possibility of overfitting. Finally, the RSSAN model achieved state-of-the-art or comparable results on three cases and can easily generalized to other remote sensing scenarios.

The future direction of our work is to exploit the attention mechanism in a 3-D-CNN-based framework, aimed to learn more discriminative spectral–spatial features for HSIC. Furthermore, how to design an end-to-end network to simultaneously implement HSI pixel classification and enhance the result of pixel classification is another research direction to study.

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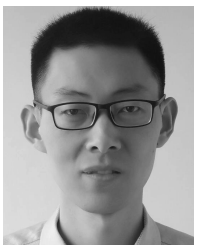
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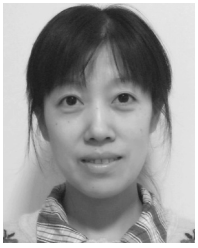
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